

SIDDHARTH PATEL

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Education

Bennett University

B.Tech in Computer Science (AI/ML) | CGPA: 9.66/10 | Dean's List (3x)

Aug 2022 – Jun 2026

Greater Noida, India

Publications

- [AICAPS '26] Hinglish Abusive Comment Detection — Hybrid mBERT/XLM-R + BiGRU; 90% accuracy on 700K+ code-mixed
- [IC3SE '25] Deep Learning-based Brain Tumor Detection — Multimodal MRI CNN; 94% accuracy with interpretability
- [MAC 2024] CNN-Based Skin Disease & Cancer Classification — IEEE DRDO Conference; 96.64% accuracy on 57 classes

Experience

AI Intern | Cograd Technologies Pvt. Ltd.

Aug 2025 – Present

- Building **multi-agent RAG systems** using **LangGraph** + **CrewAI** for adaptive learning and automated assessment
- Built **dense-retrieval RAG pipeline** for **Medha AI**; implemented LLM data pipelines for **offline-first learning**
- Performed **memory optimization** and **GGUF quantization** for on-device models; deployed **FastAPI NLP microservices** on **Azure Cloud** serving **10K+ users**

Project Developer | Govt. of India Funded Project

Oct 2025 – Present

Bennett University | Mentor: Dr. Dileep Kumar Yadav

- Building an **advanced dehazing system** for fog removal in winter to improve **road safety and visibility**
- Creating **custom fog dataset** and developing **novel architecture**; integrated **YOLO** + **ONNX** for real-time inference

Research Intern | Bennett University

Apr 2025 – Oct 2025

Mentor: Dr. Rohit Kumar Kaliyar

- Researched **PINNs** for **brain tumor growth modeling**, combining physics-based and deep learning methods
- Built **Hinglish abuse detection model** (**mBERT/XLM-R** + **BiGRU**), achieving **90% F1-score** on code-mixed text

Projects

AI-Powered Live Class Doubt Management | FastAPI, LLMs, pgvector, Redis

Dec 2025 – Present

- Built **real-time system** detecting and clustering student doubts from YouTube live chat using **pgvector**, reducing response time by **70%**
- Designed **async LLM pipeline** generating context-aware answers with **85%+ accuracy**, auto-posting consolidated responses
- Developed **Chrome extension** + **FastAPI backend** handling **100+ concurrent doubts** in large-scale live classes with intelligent prioritization

DataWhiz | FastAPI, GPT-4o, LangChain, LangGraph, Qdrant, DuckDB, LIDA, Neo4j, Azure

Sep 2025 – Jan 2026

- Architected **novel multi-agent Text-to-SQL system** handling **200+ table databases** using **GPT-4o** + **LangChain** with **vector embeddings** for semantic schema retrieval
- Implemented **multi-agent orchestration** with auto-correction and query validation; deployed on **Azure Cloud**.
- Integrated automated **visualization module** using **LIDA** for enterprise-grade analytics and data-driven decision making

Aurigen | PyTorch, Stable Diffusion XL, ControlNet, LoRA, Streamlit

Feb 2025 – Apr 2025

- Built **SDXL** + **ControlNet** pipeline with custom **LoRA fine-tuning** on curated **6K-image** jewelry dataset
- Optimized inference with **FP16 quantization** and **attention caching**; deployed via **Streamlit** for interactive design generation

Skills

Languages: Python, C++, SQL, JavaScript, java

AI/ML Frameworks: PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, Scikit-learn, XGBoost, OpenCV

GenAI & LLMs: RAG, Multi-Agent Systems, Fine-tuning, Prompt Engineering, Vector DBs (Qdrant, Pinecone, FAISS, pgvector)

Deep Learning: CNNs, Transformers, BERT, GPT, PINNs, GNNs, GANs, Diffusion Models, ControlNet, LoRA, YOLO

Backend & Cloud: FastAPI, Flask, Django, AWS, Azure, Docker, Redis, REST APIs

Databases: PostgreSQL, MongoDB, DuckDB, Neo4j, SQLite

Working on: AI Safety, Reinforcement Learning, Edge AI Deployment, Agentic Protocols

Achievements

Hackathons: Top 1% Amazon ML Challenge 2024 (30K+ teams) | Top 50 Convolve 3.0 Pan IIT (4K+ teams) | 1st Place Inspire Awards (6K+ teams) | 10th Place HackEye Hackathon (300+ teams)

Academics: Dean's List (3 semesters, top 5%) | Top 10% GATE 2025 (Data Science & AI)

Training: Amazon ML Summer School 2025 (selected among top applicants nationwide)

Certifications: IBM ML | MITx Python | NLP Specialization (Coursera) | Deep Learning Specialization (Coursera)

Leadership & Volunteering

- **Hackathon Organizer** | Cograd x IIT Bombay Techfest — Organized AI/ML hackathon, managed **1500+ participants**
- **ML Mentor** | Mentored **100+ students** during hackathons on model building, debugging, and deployment strategies
- **Technical Instructor** | Conducted workshops on **Python, ML fundamentals, and Deep Learning** for junior students